

Md Mahmudul Hasan

Graduate Research Assistant, Electrical & Computer Engineering (ECE) — Oklahoma State University
Stillwater, OK • +1 (405) 269-2869 • mahmudul.hasan@okstate.edu • [LinkedIn](#)

Summary

Ph.D. student in Electrical & Computer Engineering (Oklahoma State University) specializing in machine learning for neuroimaging and biosignals. Experienced in building end-to-end pipelines for EEG and MRI (feature engineering, deep learning, and multimodal fusion) and delivering clear, decision-ready analyses. Strong Python/PyTorch, experimental design, and result communication; interested in brain-computer interfaces and applied ML in healthcare and neurotechnology.

Education

Oklahoma State University, Stillwater, OK Ph.D., Electrical & Computer Engineering (in progress) Advisor: Dr. Qi Cheng	Aug 2025 — Present
Oklahoma State University, Stillwater, OK M.S., Electrical & Computer Engineering Focus: Machine Learning & Biomedical Signal Processing Advisor: Dr. Qi Cheng	Aug 2023 — Aug 2025 CGPA: 3.89/4.00
Military Institute of Science & Technology (MIST), Dhaka, Bangladesh B.Sc., Electrical, Electronic & Communication Engineering Advisor: Dr. Mohammed Hossam-E-Haider	Feb 2017 — Jul 2021 CGPA: 3.37/4.00

Experience

Graduate Research Assistant — ECE, Oklahoma State University Designed end-to-end ML pipelines for multimodal neuroimaging, including EEG classifiers (emotion, age, gender, user authentication, eye state) and MRI preprocessing/analysis using classical features, 2D CNNs on scalograms, and 3D CNNs on structural MRI. Proposed and implemented electrode-aware attention and gated fusion for paired EEG+MRI; set up leakage-safe subject-level cross-validation, handled class imbalance with weighted samplers, and used staged training with cosine annealing and regularization to stabilize learning on noisy time-series data. Analyzed large, complex datasets and communicated insights through decision-ready briefs, slide decks, and go/no-go recommendations for faculty and collaborators. Ensured reproducibility via deterministic seeding, clear dataset/version governance, and clean, modular NumPy/pandas/PyTorch code with experiment tracking and documentation.	<i>Stillwater, OK</i>	Aug 2023 — Present
Machine Learning Engineer — NEXT VENTURES Trained and tuned models with cross-validation and systematic error analysis to ship product features. Authored stakeholder-ready recommendation reports and model comparison summaries; implemented maintainable scikit-learn/PyTorch pipelines.	<i>Remote</i>	Jan 2023 — Jul 2023
Research Engineer — Visual Information Processing (VIP) Lab, MIST Prototyped computer-vision and HCI systems (e.g., traffic surveillance); coordinated multidisciplinary development and presented results to faculty stakeholders. Mentored junior researchers and improved documentation standards, increasing throughput and collaboration quality.	<i>Dhaka, Bangladesh</i>	Jun 2021 — Jul 2022
Engineering Intern — Dhaka Electric Supply Company (DESCO) Supported operational data analysis and real-time system monitoring to improve reliability and efficiency. Conducted field visits and drafted infrastructure reports informing process improvements and upgrade planning.	<i>Dhaka, Bangladesh</i>	Dec 2019 — Jan 2020

Selected Projects

EEG Eye-State & User Authentication — Built subject-level pipelines for eye-state detection and user ID from EEG; compared classical features vs. scalogram-based CNNs and wrote concise decision briefs on methods, results, and limitations.

EEG Emotion/Age/Gender Modeling — Developed end-to-end ML workflows for multi-task EEG prediction (emotion, age, gender), including data curation, model selection, and reporting of performance, trade-offs, and application implications.

Multimodal EEG–MRI Fusion Prototypes — Implemented electrode-aware attention and gated fusion architectures for paired EEG+MRI; evaluated leakage-safe subject-wise splits and class-imbalance handling for robust clinical decision support.

Student Performance Prediction Using Adaptive Feature Selection — Built a hybrid GA-based feature selection pipeline for educational data; compared GA, forward selection, and RFE across multiple ML models and datasets, analyzing accuracy, F1, precision/recall, feature reduction, and training time.

Real-Time Eye Gaze Estimation — Built a webcam-based gaze estimation pipeline using face/eye landmark detection and lightweight CNN regressors; optimized preprocessing and inference for low-latency tracking and stable cursor control.

Hand Gesture–Controlled HCI System — Implemented real-time hand detection and gesture classification with OpenCV/MediaPipe and classical ML, adding temporal smoothing and command mapping to control UI elements and demo devices.

Selected Publications

- *Enhanced Multi-Source Domain Adaptation for EEG Pattern Recognition via Source Selection and Distance Function Optimization*, **Asilomar Conference on Signals, Systems, and Computers**, 2025.
- *BLPnet: A new DNN model for automatic license plate detection with Bengali OCR*, **Neurocomputing**, 2022.
- *Design of CanSat for environmental monitoring and object detection*, **ICEEICT**, 2021.
- *Modelling lips-state detection using CNN for non-verbal communications*, **IC4IR**, 2021.
- *Sample-and-hold architecture with folded-cascode op-amp using gain compression (90nm CMOS)*, **ICCIT**, 2023.

Technical Skills

Programming: Python, MATLAB, C, LaTeX, HTML, Verilog.

ML/DL: PyTorch, TensorFlow, Keras, scikit-learn, FastAI; CNNs, transfer learning, time-series modeling, regression/classification.

Neurosignal/Imaging: MNE, SciPy, NiBabel, SimpleITK; time–frequency analysis; EEG/MEG/MRI preprocessing.

Computer Vision: OpenCV, MediaPipe, dlib; segmentation, detection, feature engineering.

Visualization: Matplotlib, Seaborn; publication-quality figures and concise stakeholder briefs.

Hardware/EDA: Proteus, Cadence, Quartus, PSPICE; embedded/PCB, sensors, microcontrollers.

Robotics/Embedded: Arduino, Raspberry Pi; real-time acquisition and IoT prototypes.

Leadership & Awards

- **General Secretary**, Bangladesh Student Association (OSU) 2025—Present
Lead operations, event planning, and inter-org coordination; manage budgets and communications.
- **Event Management Officer**, Muslim Students Association (OSU) 2025—Present
Plan logistics, vendor coordination, and volunteer scheduling for large-scale campus events.
- **Sports Secretary**, Bangladesh Student Association (OSU) 2024—2025
Organized multi-event competitions and community activities.
- Vice President (Organizing), MIST Innovation Club — Founding member; led workshops and events.
- Vice President (Technical Activities), MIST Robotics Club — Mentored teams and supported national competitions.
- ISO Cultural Night Champion (2023); ISO Soccer Bronze (2024,2025); multiple robotics/project awards.